Finite-Density Dynamics of Chemically Equilibrating QGP in Conformal Gubser Flow and Hard Thermal Photon Production

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Introduction

- Hydrodynamic modelling of heavy-ion collisions assumes that the quark-gluon plasma (QGP) reaches thermal and chemical equilibrium at the initial time.
- The early system is gluon dominated and quark undersaturated, so thermalisation and chemical equilibration proceed on different timescales.
- Gluon-gluon cross sections are the largest, so gluons approach saturation first. Quarks and antiquarks follow more slowly, and the medium remains chemically undersaturated with quarks lagging behind gluons.
- This paper asks how finite baryon density and transverse expansion modify chemical equilibration and, consequently, hard thermal-photon emission.

Bjorken versus Gubser flow

- **Limitation of 1D models:** Bjorken flow describes longitudinal, boost-invariant expansion but neglects transvers dynamics
- **The Gubser flow:** adds an azimuthally symmetric radial expansion while retaining boost invariance and a semi-analytic conformal framework

Property	**Bjorken Flow**	**Gubser Flow**
Expansion	Longitudinal only	Longitudinal + radial
Symmetry	Boost-invariant	Boost-invariant + Conformal $SO(3)$
Transverse profile	Homogeneous	Conformal fluids only

Weyl Rescaling and Hydrodynamic Simplification

- The expanding QGP in Milne coordinates $x^\mu = (\tau, r, \phi, \eta_s)$ is mapped to de Sitter coordinates $(\rho, \theta, \phi, \eta_s)$ through the Weyl rescaling $d\hat{s}^2 = ds^2/\tau^2$.
- In de Sitter coordinates, the fluid four-velocity is static, $\hat{u}^\mu = (1, 0, 0, 0)$; the thermodynamic fields still evolve with de Sitter time ρ.
- The map reduces the coupled hydrodynamic and chemical evolution from PDEs in (τ, r) to ODEs in ρ:

$$d\hat{s}^2 = d\rho^2 - \cosh^2\rho(d\theta^2 + \sin^2\theta d\phi^2) - d\eta_s^2$$

Partonic Phase-Space and Chemical Non-Equilibrium

- Chemical fugacities λ_i quantify deviations from equilibrium: $\lambda_i = 1$ is chemically saturated, while $\lambda_i < 1$ is undersaturated.
- The baryon-density dependence enters through $x_q = \mu_q / T$. Defining $\lambda_{Q,\bar{Q}} = e^{\pm x_q} \lambda_{q,\bar{q}}$ gives

$$f_{q,\bar{q}} = \frac{\lambda_{Q,\bar{Q}}}{\lambda_{Q,\bar{Q}} + e^{\beta(u \cdot p)}}.$$

- The subsequent rate calculation uses factorized thermal distributions, and the dominant chemical reactions are
  - $gg \leftrightarrow ggg$ for gluon multiplication,
  - $gg \leftrightarrow q\bar{q}$ for quark–antiquark production.

Coupled hydrodynamic and chemical evolution

- Energy conservation couples the temperature to the evolving quark and gluon fugacities:

$$\frac{4\hat{T}'}{\hat{T}} + \frac{a_2\hat{\lambda}'_g + b_2(\hat{\lambda}'_Q + \hat{\lambda}'_{\bar{Q}})}{a_2\hat{\lambda}_g + b_2(\hat{\lambda}_Q + \hat{\lambda}_{\bar{Q}})} + \frac{8}{3}\tanh\rho = 0.$$

- The master equations combine dilution from expansion with the inelastic reaction rates $\hat{R}_2$ and $\hat{R}_3$:

$$\frac{3\hat{T}'}{\hat{T}} + \frac{\hat{\lambda}'_Q}{\hat{\lambda}_Q} + 2\tanh\rho = \hat{R}_2\frac{a_1}{b_1}\left(\frac{\hat{\lambda}_g}{\hat{\lambda}_Q} - \frac{\hat{\lambda}_{\bar{Q}}}{\hat{\lambda}_g}\right).$$

Initial conditions used for the main results

$\tau_0 = 0.23\ \mathrm{fm}, \quad T_0 = 0.83\ \mathrm{GeV}, \quad \lambda_q^0 = 0.03, \quad \lambda_g^0 = 0.14, \quad x_q^0 \equiv \mu_q^0/T_0 = 0,\ 0.5,\ 1.0,\ 1.5.$

Proper-time evolution of the plasma

- Mapping back to Milne coordinates using, $T = \hat{T}/\tau$ and $\mu_q = \hat{\mu}_q/\tau$, gives the physical proper-time evolution of T, λ_q, λ_g, and μ_q at the fireball center.
- At the fireball centre, increasing x_q^0 lowers λ_g at freeze-out from 0.76 to 0.72 and raises λ_q from 0.34 to 0.35.

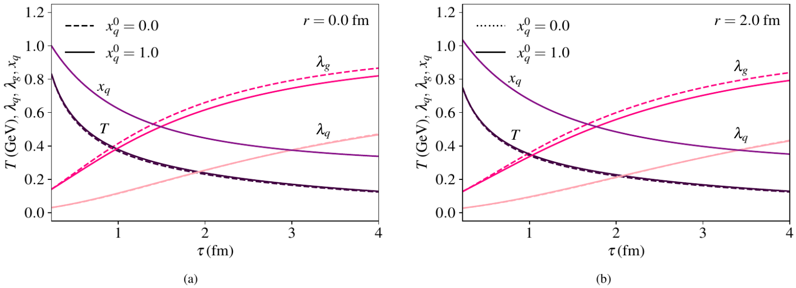
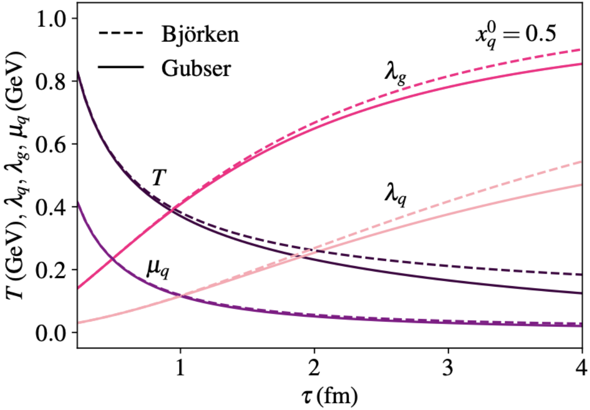


Fig. 8 — Naik & Sreekanth, arXiv:2606.31749v1.

Transverse Expansion Delays Chemical Saturation

- The figure compares Gubser and Bjorken evolution at $r = 0$ for $x_q^0 = 0.5$.
- Radial expansion produces faster cooling in Gubser flow.
- Consequently, λ_q and λ_g remain below their Bjorken values during the evolution.



Solid: Gubser flow; dashed: Bjorken flow (paper Fig. 14).

Evolution of the equilibration rates

- The figure shows the scaled quark-production rate $R_2/T \approx 0.025$ and gluon-multiplication rate $R_3/T \approx 0.20$.
- The gluon channel is initially about eight times faster. This is the origin of the quark lag.
- During the evolution, R_2/T increases while R_3/T decreases, and finite density affects the two rates differently.

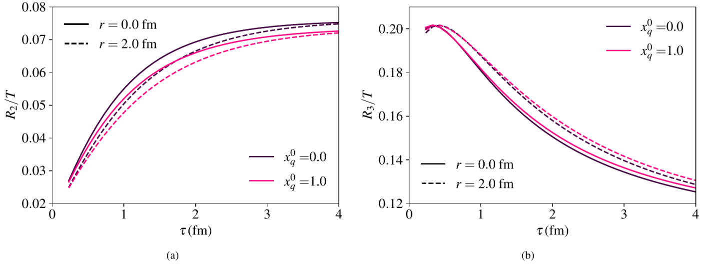
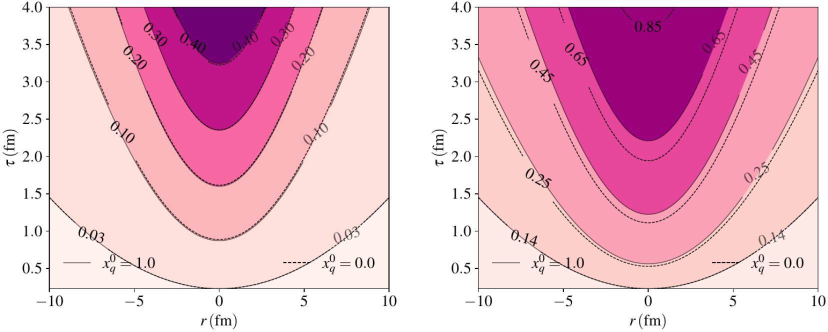


Fig. 9, rates scaled by temperature — Naik & Sreekanth, arXiv:2606.31749v1.

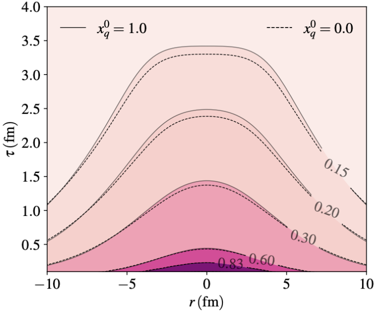
Evolution of Chemical Saturation

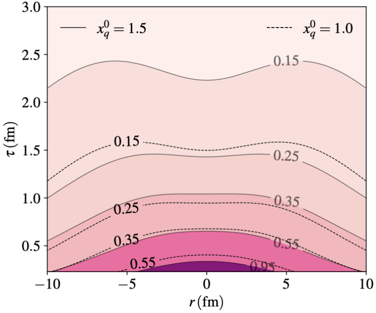
- The r–τ contours show the spatial dependence of λ_q and λ_g generated by radial expansion.
- The central region reaches larger fugacities, while the cooler peripheral region remains more strongly undersaturated and freezes out earlier.



Temperature and Chemical-Potential Profiles

- Temperature and quark chemical potential develop nonuniform profiles in the r–τ plane.
- Solid curves show finite-density evolution; dashed curves show the zero-density comparison. The hottest and densest region remains near the fireball center.





Thermal photons as a probe

- **Electromagnetic coupling:** thermal photons interact only electromagnetically.
- **No rescattering:** once produced they leave the fireball without further interaction.
- **Emission history:** the yield integrates over the entire space-time evolution, so the spectrum carries information about the conditions at each emission point rather than about the final state alone.

Photon production: processes and rates

- **One-loop processes:** quark-antiquark annihilation and Compton scattering.
- **Two-loop processes:** bremsstrahlung, which dominates at low p_T, and annihilation with scattering (AWS), which dominates elsewhere.
- **Non-equilibrium rates:** the same factorisation used in the rate equations is applied to the photon rates,

$$f_1 f_2(1 \pm f_3) = \lambda_1\lambda_2\lambda_3\, f_1^{eq} f_2^{eq}(1 \pm f_3^{eq}) \;+\; \lambda_1\lambda_2(1-\lambda_3)\, f_1^{eq} f_2^{eq}.$$

The first term carries the product of all participating fugacities; the second is the quantum blocking correction, which survives only out of equilibrium.

- At fixed temperature, chemical undersaturation reduces the production rate for all four processes. For $\lambda_g = 2\lambda_q = 0.6$ and $x_q = 0.5$ the one-loop contribution falls by roughly 50% at $E = 1$ GeV and 78% at 3.5 GeV, and the two-loop contribution by about 82% across the range.

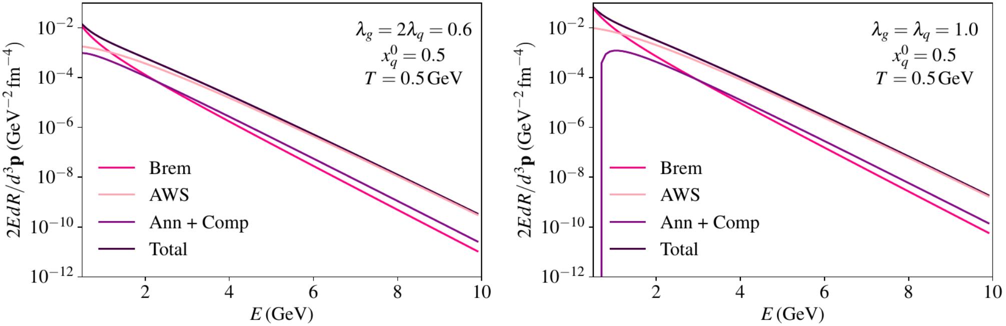


Fig. 1, non-equilibrated (left) and equilibrated (right) rates at $T = 0.5$ GeV — Naik & Sreekanth, arXiv:2606.31749v1.

Photon spectra from the expanding fireball

- The solid curves describe chemical evolution; dashed curves use the fully saturated benchmark.
- Fewer quarks and gluons are available, so the local emission rate drops everywhere.
- The space-time integrated thermal photon yield is suppressed by roughly two orders of magnitude.
- In high p_T the suppression is strongest because yield is dominated by the hot early stage in which the medium is furthest from saturation.

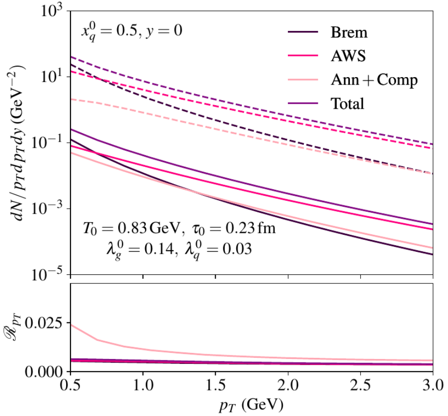


Figure: Photon spectra, $x_q^0 = 0.5$

Fig. 15 — Naik & Sreekanth, arXiv:2606.31749v1.

Early-time fraction of the yield

- The fraction of photons of given p_T emitted before a proper time τ_*,

$$\mathcal{Y}_*(p_T; \tau_*) = \frac{(dN/p_T dp_T dy)_{\tau<\tau_*}}{(dN/p_T dp_T dy)}.$$

- Out of equilibrium, high-p_T photons reach a large fraction of their total by $\tau_* \simeq 1$ fm; the equilibrated case needs about 2 fm.

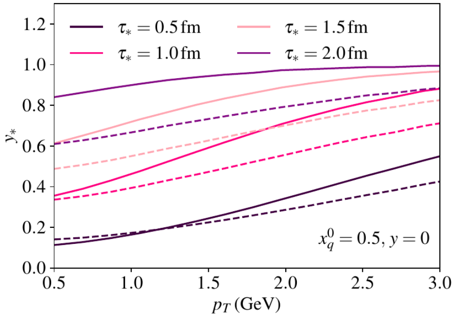


Figure: Cumulative early-time fraction $\mathcal{Y}_*$

Fig. 17 — Naik & Sreekanth, arXiv:2606.31749v1.

Summary and Conclusions

- Finite density and transverse expansion delay chemical saturation; the system reaches freeze-out with $\lambda_q, \lambda_g < 1$.
- Chemical undersaturation suppresses the hard thermal-photon yield relative to the equilibrated benchmark.
- At the same time, the normalized early-time fraction of high-p_T emission increases.
- The quantitative result depends on ideal conformal Gubser flow, extreme initial undersaturation, factorized photon rates, and the omission of prompt, pre-equilibrium, and hadronic photon sources.